

Begum Akkas

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EDUCATION

The University of Chicago (Honors)

Master of Science in Computational Analysis and Public Policy

Chicago, IL

June 2026

The George Washington University (Summa Cum Laude)

Bachelor of Arts in Economics and International Affairs, Security Policy Concentration

Washington, DC

May 2021

SKILLS

Technologies: Python, SQL (MySQL, DuckDB), JS (D3), HTML/CSS, R, ArcGIS, Git/GitHub, AWS (S3, EC2, SNS/SQS), Ruby on Rails, Svelte

Libraries: PyTorch, Pandas, GeoPandas, NumPy, scikit-learn, Flask (REST APIs), Matplotlib, Altair, Boto3, Plotly, Shapely, Pytest

Languages: English (native); Turkish (native)

PROFESSIONAL EXPERIENCE

Fraym

Data Science Intern

Chicago, IL

June 2025 – August 2025

- Built and deployed a Streamlit dashboard providing the DS, Engineering, and Product teams visibility into ensemble ML model performance across jobs, models, and geographies; features include RMSE distributions (KDE + histogram), outlier detection, model/feature frequency analysis, and cross-fold comparison.
- Engineered a configurable data sampling pipeline integrating AWS S3 and an internal database, enabling users to filter by submission ID, country, target variable, and time frame to pull representative data subsets for analysis.
- Identified performance gaps in the ensemble model through RMSE outlier analysis, contributing to an estimated 5–10% improvement in model accuracy; authored a technical proposal outlining project architecture, data flow, and interpretability methods for the DS team.
- Produced end-to-end technical documentation including a user guide covering dashboard setup, data fetching, and feature walkthroughs to support adoption across teams.

Berkeley Research Group

Consultant, Senior Associate, Associate

Washington, DC

August 2021 – May 2024

- Designed and maintained SQL analytics pipelines to process large-scale datasets for 2–3 concurrent healthcare litigation matters, delivering data-driven analyses and expert reports for clients and counsel.
- Independently learned Snowflake to process millions of records under a two-week deadline on a pro bono case; presented findings directly to the client as the most senior staff member on the engagement.
- Authored two expert reports summarizing quantitative analyses and recommendations for senior leadership and outside counsel; managed 4–5 junior analysts across 2–3 simultaneous cases.

PROJECTS

Green Space Accessibility in Chicago | Python, GeoPandas, OpenStreetMap API

- Built a geospatial accessibility index for ~500 affordable housing units, combining spatial buffering and fuzzy name matching (Jaro-Winkler) to link park polygons to Yelp/Google ratings data.
- Designed a composite scoring methodology weighting parks by size and review-adjusted quality, normalized across the full housing dataset; revealed disparities in green space access across Chicago neighborhoods.

Hottest U.S. Songs | Python, D3.js, HTML/CSS, Cloudflare

- Cleaned and analyzed 60+ years of Billboard #1 chart data to surface trends in artist demographics and songwriting credits; found women are substantially more represented as performers than as credited songwriters.
- Built and deployed an interactive D3 visualization enabling users to explore chart history and demographic patterns.

Homework Tracker App | Ruby on Rails, HTML/CSS, PostgreSQL, OpenAI API

- Built a full-stack web app that ingests PDF syllabi via the OpenAI API to extract structured assignment data (due dates, priority, exam flags) into a sortable task dashboard, eliminating manual data entry.